Two-way tables

1 50 students were asked whether or not they were allergic to nuts and dairy. The two-way table is provided below:

	Allergic to Nuts	Not Allergic to Nuts
Allergic to Dairy	6	11
Not Allergic to Dairy	6	27

a How many students are allergic to nuts?

b How many students are allergic to nuts or dairy, or both?

c How many students are allergic to at most one of the two things?

2 Some students were asked if they are left or right handed. The results are provided in the following table:
A student is picked from this group at random.

	Left-handed	Right-handed	Total
Female	10	39	49
Male	5	50	55
Total	15	89	104

a How many of the students are right-handed males?

b How many of the students are left-handed?

c How many of the students are not right-handed males?

3 Mr. Tobit asked the students in his class to pick their favourite subject. He displayed the results in the following two-way table:

	Boys	Girls
Maths	11	12
Music	8	18
Science	13	11
English	15	10

a How many girls did not pick maths as their favourite subject?

b How many students picked music?

c How many boys are in Mr. Tobit's class?

4 170 tennis players were asked whether they would support equal prize money for the women's and men's draw.
Complete the following table:

	Support	Do not support
Males		35
Females	71	15

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5 A healthy living initiative asked people to describe how often they go to the gym. Their responses are shown in the following table:

	Female	Male
Frequently	35	37
Rarely	35	13

a How many people were surveyed?

b If one person is chosen at random, find the exact probability that they are a male who frequently attends the gym.

c If one person is chosen at random, find the probability that they attend the gym rarely.

6 Members of a gym were asked what kind of training they do. The two way table shows the results:

	Cardio	Weight
Male	11	30
Female	47	12

a How many gym members were asked altogether?

b How many members do weight training?

c If a member is chosen at random, what is the probability that they do weight training?

d According to the table which is more likely doing weight training, men or women?

7 A nonsmoking initiative asked smokers to describe how often they smoke:

	Female	Male
Frequently	57	34
Rarely	35	14

a How many people were surveyed?

b If one person is chosen at random, what is the probability that they are a frequent male smoker?

c If one person is chosen at random, what is the probability that they smoke rarely?

8 A town has two campsites to choose between which both offer tent and cabin accomodation. This two-way table records the campers choices in one summer:

	Tent	Cabin
Sunny Campground	178	62
Platypus Creek	101	281

a How many people stayed in one of the town's campsite?

b State the proportion of the people that stayed in a tent.

c State the proportion of the people who stayed in cabins was in Sunny Campground. .

d State the proportion of people stayed at Platypus Creek.

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9 A group of tourists were asked whether they spoke Mandarin or Spanish.

a Complete the following table:

b How many people speak both languages?

c If one person is chosen at random, find the probability that they speak neither language.

d If one person is chosen at random, find the probability that they speak only one of the languages.

	Spanish	Not Spanish	Total
Mandarin	58	10	
Not Mandarin	15	17	

10 Sophia asked some people in her community whether they were vegetarian or not. 29 responders said they were vegetarian, of which 8 were children. 14 children said they were not vegetarian, and 11 adults said they are not vegetarians.

a Construct a two-way table based on the results of Sophia's survey.

b State the proportion of responders that are vegetarian.

c State the proportion of adults that are vegetarian.

11 The following table shows the number of trains arriving either on time or late at a particular station:

a How many trains were late on Friday?

b How many trains passed through the station on Wednesday?

c How many trains were on time throughout the entire week?

d State the proportion of trains that were on time over the whole week. .

	Arrived on time	Arrived late
Monday	23	9
Tuesday	20	5
Wednesday	27	8
Thursday	28	14
Friday	15	12
Saturday	22	6
Sunday	26	13

12 In a study, some people were asked whether they lie. A partially completed two-way table of the results is shown below.

a Complete the following table:

b Of those in the study, one is chosen at random. Find the probability that they said they never lie.

	Lie	Dont Lie	Total
Children	15		25
Adults	10		
Total			60

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13 This table describes the departures of trains out of a train station for the months of May and June:

	Departed on time	Delayed
May	123	32
June	124	47

a How many trains departed during May and June?

b State the proportion of the trains in June that were delayed. Write your answer as a percentage to one decimal place.

c State the proportion of the total number of trains during the 2 months that were ones that departed on time in May. Give your answer as a percentage rounded to one decimal place.

d Find the probability that a train selected at random in June would have departed on time. .

e Find the probability that a train selected at random from the 2 months was delayed.

14 At a local university, students were asked what their favourite subject at high school was and what they have decided to major in after 3 years of university. The results are shown in the following table:

	Maths favourite	Science favourite	Music favourite	Art favourite	Total
Maths major	76	20	61	43	200
Science major	64	46	53	59	222
Music major	64	11	67	59	201
Art major	6	19	38	74	137
Total	210	96	219	235	760

One student is chosen at random. Find:

a The probability that a student's favourite subject was mathematics at high school.

b The probability that a student is majoring in music or arts at university.

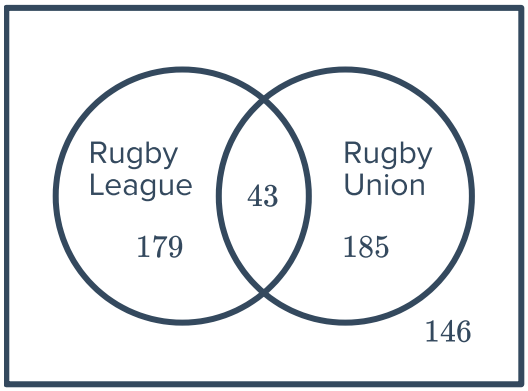
c The probability that a student's favourite subject was music and they studied something different at university.

d The probability that a student's major is the same as their favourite subject?

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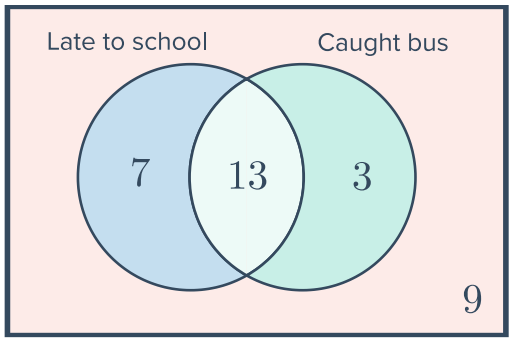
Two-way tables and Venn diagrams

15 Consider the Venn diagram:
Complete the table of values below.

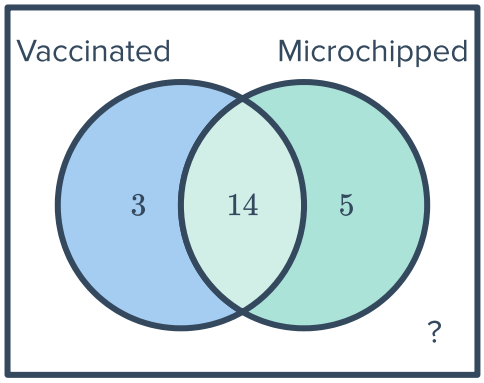


	Play Rugby League	Don't play Rugby League
Play Rugby Union		
Don't play Rugby Union		

16 A student makes a Venn diagram of students who are late to school, and students who catch the bus to school.
Construct a two-way table based on the Venn diagram.

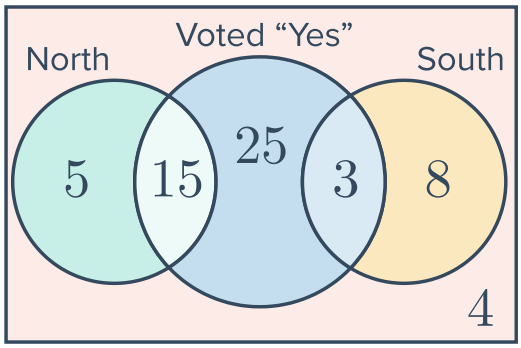


17 A vet has 28 pets visit their practice in a day.
The pets are categorised based on whether they have been vaccinated and whether they have been microchipped.
Construct a two-way table based on the Venn diagram.



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18 60 residents of a city were asked "Do you support the construction of the new train station? ". The residents questioned were also classified as living in the north, south or in the inner city.
Construct a two-way table based on the Venn diagram.



19 Students in Irene's class were asked if they owned a dog and asked if they owned a snake. The following two-way table shows that information:

	Owns a dog	Doesn't own a dog
Owns a snake	2	3
Doesn't own a snake	13	11

Construct a Venn diagram that represents the information provided in the two-way table.

20 100 random people in Australia were surveyed, examining their carbon footprint and the city they lived in. The people were then categorised as living in either an urban or regional location, and whether that person has a high carbon emission or low carbon emission.

	Urban	Regional	Total
High Emission	37	13	50
Low Emission	24	26	50
Total	61	39	100

Construct a Venn diagram that represents the information provided in the two-way table.

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